Rahul Jogadhenu

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Education

University of California - Los Angeles (UCLA) Los Angeles, CA
Bachelor of Science in Computer Science & Engineering | GPA: 3.44/4.0 *Expected June 2028*
Relevant Coursework: Data Structures, Algorithms, Software Construction, Operating Systems, Machine Learning, Computer Graphics, Networking, Computer Systems Security, Systems & Signals, Digital Design Lab

Technical Skills

Languages: C++, C#, Python, Java, HTML/CSS, JavaScript, Bash, MySQL, Verilog, HLSL, GLSL
Frameworks: React.js, React Native, Express.js, Node.js, Expo, OpenSSL, Three.JS
Developer Tools: Unity, Unreal Engine, Git, GitHub, Cursor, VS Code, Jenkins, FPGAs, Jira, Confluence, Bitbucket
Technologies: REST APIs, Linux, Database Design, Networking, Software Testing, CI/CD pipelining, UML

Experience

Software Engineering Intern June 2026 – Sep. 2026
Werfen *San Diego, CA*
- Developed a Python and bash based internal DevOps package which uses a device-side fuzzing based approach to test embedded medical-device software on multiple products, automating bug detection, and deployment validation
- Refactored 8,000+ lines of Python code into a centralized repository by leveraging Cursor for AI assisted coding and debugging in a Linux development environment, optimizing code and eliminating module duplication
- Automated CI/CD build document generation pipelines using Jenkins, streamlining compliance tracking
- Collaborated in Agile Scrum sprints, executing bug fixes, testing, and software verification while working cross-functionally to support on-time releases

Undergraduate Researcher Oct. 2025 – Present
UCLA Health Cross Movement Disorders Lab *Los Angeles, CA*
- Architected and optimized virtual reality environments in Unity and Unreal Engine for clinical studies on freezing of gait in Parkinson's disease patients, controlled testing and data collection
- Implemented EEG patient data logging and LSL network integration for synchronized real-time data collection

Course Instructor Sep. 2025 – Present
UCLA Samueli School of Engineering *Los Angeles, CA*
- Instructed game development fundamentals, C# object-oriented programming, and Agile game development workflows to over 40 undergraduate students using the Unity engine

Game Jams Chair Oct. 2025 – Present
ACM Studio at UCLA *Los Angeles, CA*
- Organized the 2024 and 2025 Fiat Ludum Game Jams which had over 100 participants and industry judges
- Leading a team of over 15 developers to develop the video game Cloudy Critters, focusing primarily on programming architecture, system design, and gameplay balancing

Projects

AI Tour Guide App | *React Native, Expo, Express.js, MongoDB* Oct. 2025 – Dec. 2025
- Developed a Tour Guide mobile app app which provides real-time audio based information about nearby locations by leveraging the Gemini and Google Maps APIs, using the Expo framework with React Native
- Engineered the application backend, building the accounts system, database storage, and 10+ REST API endpoints using Express.js alongside MongoDB
- Used React Native and CSS to design the app's login and location history pages

Ext2 File System | *C, Linux* Nov. 2025
- Engineered a custom C-based ext2 file system capable of mounting, reading and writing files, supporting symbolic links, and seamlessly integrating with the standard Linux Virtual File System (VFS)
- Simulated object-oriented design patterns utilizing C structs and function pointers to architect low-level operating system components and implement robust manual memory management

Verilog Breakout | *Verilog, Xilinx Vivado, Basys3 FPGA* Feb. 2026 – Mar. 2026
- Programmed the arcade video game Breakout using Verilog for an FPGA board, utilizing a VGA display for graphics alongside the Basys3 board for input and processing
- Implemented functionality such as rendering, frame synchronization, input, clock dividers, and scoring logic